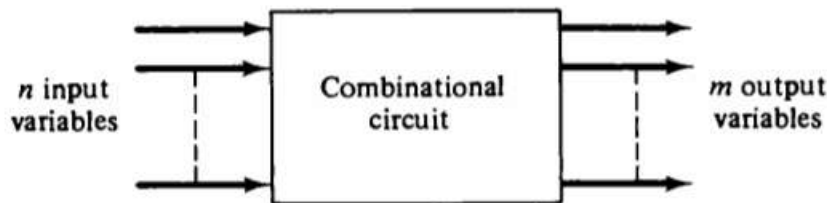
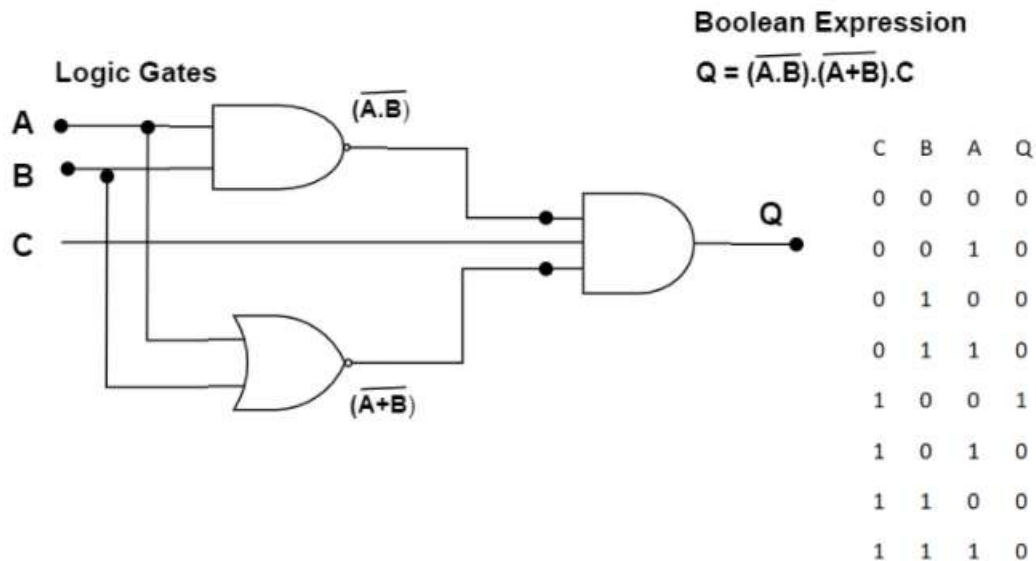


Computer System Architecture
COMP201TH
Lecture-7
Combinational Circuits

- Combinational Circuits:
 - A combinational circuit is a connected arrangement of logic gates with a set of inputs and outputs.
 - The behaviour of a combinational circuit is memoryless i.e. the output depends solely on its most recent input and is independent of the circuit's past history.
 - In general term, a combinational circuit consists of n binary inputs and m binary outputs.
 - Combinational Logic circuits are made up from basic logic NAND, NOR or NOT gates that are combined or connected together to produce more complicated switching circuits.
 - Any combinational circuit can be implemented with only NAND and NOR gates as these are classified as “universal gates.”

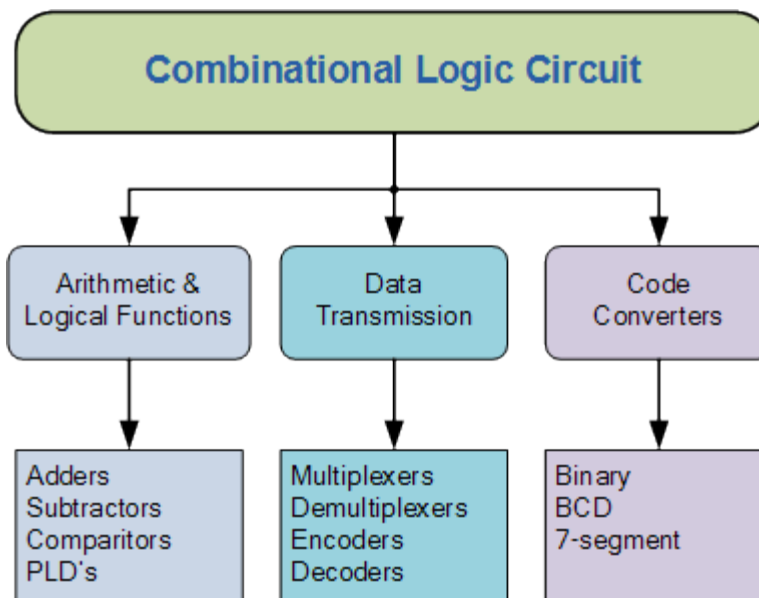


- As with logic gates, a combinational circuit can be defined in three ways:
 - Truth Table: For each of the 2^n possible combinations of input signals, the binary value of each of the m output signal is listed.
 - Graphical Symbols: The interconnected layout of gates is depicted.
 - Boolean Equations: Each output signal is expressed as a Boolean function of its input signals.



Representation of Combinational Logic Circuits

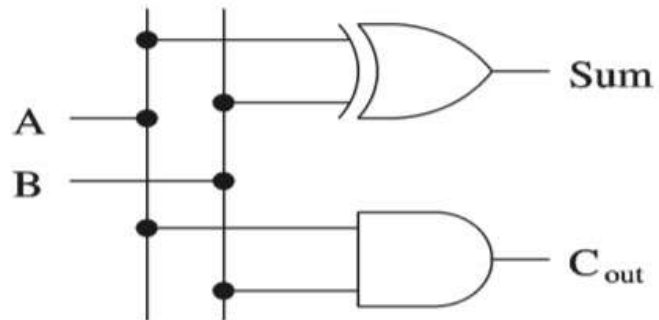
- Common combinational circuits made up from individual logic gates that carry out a desired application include Multiplexer, De-multiplexers, Encoders, Decoders, Full and Half Adders etc.
- Classification of Combinational Logic:



- **Half Adder:**

- The most basic digital arithmetic circuit is the addition of two binary digits.
- A combinational circuit that performs the **arithmetic addition of two bits** is called a half-adder.

A	B	Sum	C _{out}
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

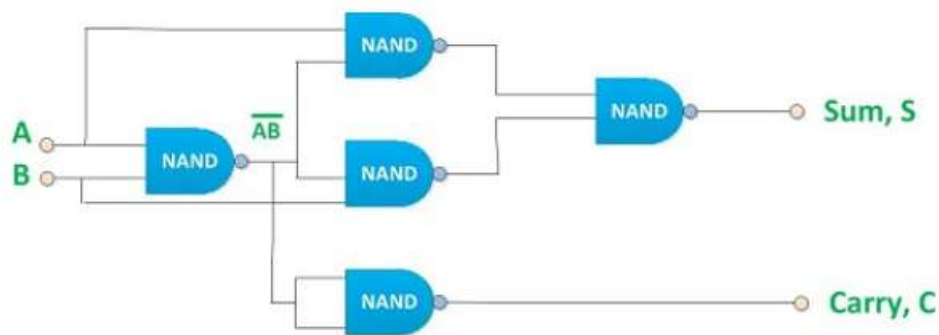


Half-adder truth table and implementation

$$S = A \oplus B = A \oplus B$$

$$C = AB$$

- To implement half adder using NAND gates; we require 5 NAND gates.



Half Adder Using NAND Gates

$$0+1 = 1 + 0 = 1 \rightarrow S=1, C= 0$$

$$1+1 = 10 \rightarrow S=0, C=1$$

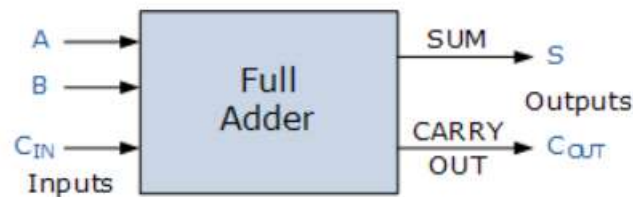
$$1+1+1 = 11 \rightarrow S=1, C=1$$

- **Limitations of Half Adder:**

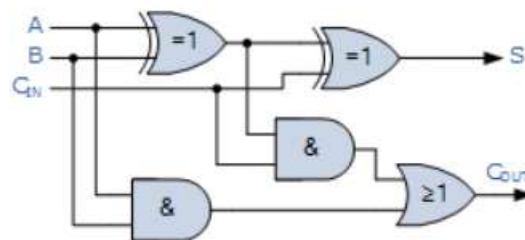
- The reason it's called half adder is that there is no scope to add the carry bit from previous bit. This is a major limitation of half adders when used as binary adders especially in real time scenarios which involves addition of multiple bits.
- To overcome this limitation, full adders are developed.

- **Full Adder:**

- A full-adder is a combinational circuit that performs the **arithmetic sum of three input bits**.

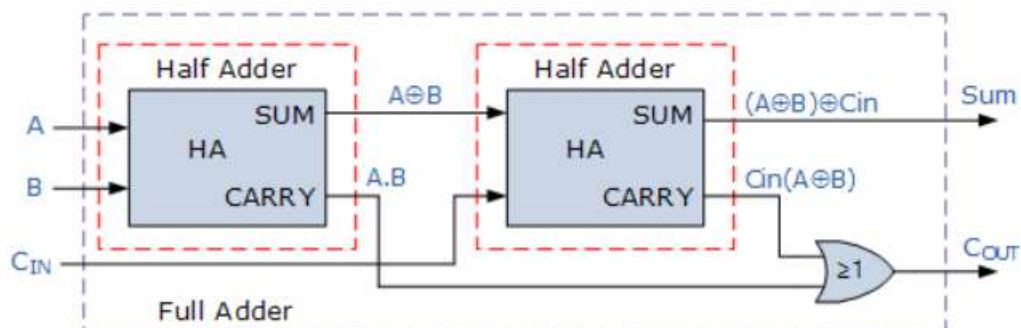


Block Diagram of Full Adder



Logic Diagram of Full Adder

- The full adder can be thought of as two half adders connected together, with the first half adder passing its carry to the second half adder.



○

- Truth Table for Full Adder is:

Truth Table				
C-in	B	A	Sum	C-out
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

- The Boolean Expression for a full adder is:

$$\text{Sum} = (A \oplus B) \oplus C_{in}$$

$$\text{C-Out} = A.B + C_{in}(A \oplus B) = AB + AC_{in} + BC_{in}$$

$$\mathbf{AB + AC_{in} + BC_{in}}$$

$$= AB + AC_{in} + BC_{in}(A + A')$$

$$= ABC_{in} + AB + AC_{in} + A'BC_{in}$$

$$= AB(1 + C_{in}) + AC_{in} + A'BC_{in}$$

$$= AB + AC_{in} + A'BC_{in}$$

$$= AB + AC_{in}(B + B') + A'BC_{in}$$

$$= ABC_{in} + AB + AB'C_{in} + A'BC_{in}$$

$$= AB(1 + C_{in}) + AB'C_{in} + A'BC_{in}$$

$$= AB + AB'C_{in} + A'BC_{in}$$

$$= AB + C_{in}(AB' + A'B) = \mathbf{AB + C_{in}(A \oplus B)}$$